$$\begin{aligned} \langle x|p\rangle &= \frac{1}{\sqrt{2\pi}} e^{\frac{ipx}{\hbar}} \\ \langle p|x\rangle &= \frac{1}{\sqrt{2\pi}} e^{\frac{-ipx}{\hbar}}. \end{aligned} \qquad (8.18)$$

The second equation is simply the complex conjugate of the first. These results are easy to verify if you keep in mind that |*x*> is represented by a delta function. I'd like to mention two important points before moving further:

1. Eq. 8.17 represents a momentum eigenfunction in the position basis. In other words, although it represents a momentum eigenstate, it is a function of *x*, and *not* an explicit function of *p*.

2. We've been using the symbol ψ for both position and momentum eigenstates. A mathematician might not approve of using the same symbol for two different functions, but physicists do it all the time. $\psi(x)$ is just the generic symbol for whatever function we happen to be discussing.

At this juncture, we begin to get a glimmer of why the wave function is called the wave function. What you should notice is that the eigenfunctions (wave functions representing eigenvectors) of the momentum operator have the form of waves—sine waves and cosine waves, to be precise. In fact, we can now see one of the most fundamental aspects of the wave-particle duality of quantum mechanics. The wavelength of the function

$$e^{\frac{ipx}{\hbar}}$$

is given by